

AAYUSH NITIN MAHESHWARKAR

Senior Software Engineer · Java · Spring Boot · Microservices · Distributed Systems · Agentic AI
aayushmaheshwarkar@gmail.com · [linkedin.com/in/aayush-nitin-maheshwarkar](https://www.linkedin.com/in/aayush-nitin-maheshwarkar) · aayushnm.github.io · Orlando, FL

PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience contributing to large-scale, revenue-generating enterprise platforms. Skilled in Java, Python, Spring Boot, and REST APIs, with a strong track record of cross-functional delivery of complex, high-stakes features across global systems. Hands-on experience in Agentic AI instruction design, production hardening, and the full software development lifecycle.

PROFESSIONAL EXPERIENCE

Senior Software Engineer · Federal Express Corporation

Jul 2021 – Present

Orlando, FL · Promotion: Full Stack Developer II (Jul 2021) → Full Stack Developer III (Feb 2023)

- Engineered Java and Spring Boot cloud-native rating microservices processing 2.5M+ peak daily requests, including the Quote Platform pricing engine (\$450M billed / \$1.8BN quoted across 69 markets) and a surcharge calculation engine exposed via REST APIs generating \$150M+ in annualized revenue, contributing to \$600M+ in total platform revenue.
- Implemented precedence logic in Java and Spring Boot for the dangerous-goods surcharge category, governing how surcharges combined across country-specific rules and enabling migration of 34,000+ enterprise customers, unlocking \$41M direct and \$1.2B in linked revenue.
- Reduced customer-facing rate-quote API latency by ~600ms by designing proactive Okta JWT pre-caching using Caffeine with async background refresh, eliminating synchronous token-fetch overhead from the pricing API.
- Owled quality standards and final approval authority for Microsoft Copilot LLM instruction sets across multiple enterprise platforms, establishing evaluation criteria for clarity, completeness, security, and accuracy before production release.
- Contributed to a LangChain and Claude-powered agentic pipeline eliminating a 5-day manual error-correction cycle; implemented database-level retry controls and authored Python scripts for bulk data cleanup and revenue recovery.
- Implemented Caffeine Loading Cache across rating services by identifying high-value cache candidates, reducing downstream service calls and improving pipeline response times across 2.5M+ daily requests.
- Participated in Agile Scrum ceremonies, production incident triage, and peer code reviews across cross-functional teams spanning engineering, product, and business stakeholders.

Selected Work Projects

Agentic AI Error-Correction Pipeline - LangChain + Claude

- Contributed to a LangChain and Claude-powered agentic pipeline that autonomously triages and remediates functional rating failures before invoicing resubmission, built to handle a projected 10x surge from 2,000 to 20,000 daily shipments and eliminate a 5-day manual remediation cycle.
- Implemented database-level retry controls to prevent infinite loops on unresolvable errors and authored Python automation for bulk data cleanup, accelerating recovery of previously uninvoiced revenue.

Rate Quote API Latency Optimization - Caffeine + Okta JWT

- Diagnosed synchronous Okta JWT token-fetch as the dominant bottleneck on the customer-facing rate-quote API processing 2.5M+ daily requests.
- Designed and implemented Caffeine refreshAfterWrite pre-caching with async background refresh, eliminating synchronous token overhead and reducing API latency by ~600ms; pattern adopted by adjacent teams.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Shell Scripting, C

Backend & Microservices: Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST APIs, Feign, Resilience4j (Circuit Breakers, Retry), JMS Messaging, Service Discovery (Eureka), Cloud-Native Architecture, Event-driven Architecture

Cloud & DevOps: Docker, Jenkins CI/CD, Gradle, Linux, Git

Security: OAuth 2.0, SSO, Okta, JWT, Spring Security

Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Caffeine (Loading Cache, refreshAfterWrite)

Observability: Splunk, Dynatrace, Grafana, AppDynamics

AI & LLMs: Agentic workflow design, LangChain, Claude, Microsoft Copilot, LLM instruction engineering, prompt evaluation, RAG fundamentals, AI governance, guardrails, human-in-the-loop review

Practices: Agile (Scrum, Kanban), Unit Testing (JUnit), Code Reviews, Production Incident Triage

ACHIEVEMENTS

- FedEx Hall of Fame Award (FY24):** Recognized for contributions to the Quote Platform initiative - \$450M billed / \$1.8B quoted across 69 global markets in FY23.
- AAAI 2020 Fall Symposium on AI for Social Good (Co-Author):** "Socioeconomic and Geographic Variations Impacting the Spread of Malaria." [\[GitHub\]](#)

EDUCATION

M.S. Information Technology

Rensselaer Polytechnic Institute · Troy, NY · CGPA 3.8/4.0

Aug 2019 – Dec 2020

B.Tech Electronics & Communication Engineering (Minor: Computer Science)

PES University · Bengaluru, India · CGPA 7.63/10 · Minor CGPA 8.95/10

Aug 2015 – Jul 2019